

*MA.7.AR.1.1***Benchmark****MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.**

Example: $(7x - 4) - \left(2 - \frac{1}{2}x\right)$ is equivalent to $\frac{15}{2}x - 6$.

Benchmark Clarifications:

Clarification 1: Instruction includes linear expressions in the form $ax \pm b$ or $b \pm ax$, where a and b are rational numbers.

Clarification 2: Refer to Properties of Operations, Equality and Inequality (Appendix D).

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.1, MA.7.NSO.2.2

Terms from the K-12 Glossary

- Coefficient
- Linear Expression
- Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.6.AR.1.4

Next Benchmarks

- MA.8.AR.1.2, MA.8.AR.1.3

Purpose and Instructional Strategies

In grade 6, students generated equivalent algebraic expressions with integer coefficients. In grade 7, students perform operations (adding and subtracting) on linear expressions with rational coefficients and multiple variables. In grade 8, students will multiply two linear expressions with rational coefficients as well as continuing to write equivalent expressions.

- Students extend their understanding of operations with rational numbers to linear expressions. This will be critical to solving equations and inequalities having more than one step. This is also an opportunity to build fluency within MA.7.NSO.2 (*MTR.3.1*).
- Use manipulatives such as algebra tiles to emphasize the difference between linear and constant terms for integers. Once clear and confident, the transition from manipulatives to the abstract to include rational numbers (*MTR.2.1*).
 - Algebra Tiles

$$(7x - 4) + \frac{1}{2}x - 2 = \frac{15}{2}x - 6$$

- Variables are not limited to x . Instruction includes using various lowercase letters for their variables; however, o , i and l should be avoided as they too closely resemble zero and one.

- Instruction includes students working within the same type of rational numbers when appropriate.
 - For example, if students are given fractions, the solution should be demonstrated in fractions and not be converted to decimals.

Common Misconceptions or Errors

- Students may incorrectly add and subtract terms that are unlike. To address this misconception, begin with a sorting activity to organize terms into groups before performing operations with those terms (*MTR.2.1*). Have students verbalize the criteria for being considered like terms to ensure they are focused on the correct similarities.
- Students may incorrectly distribute the negative sign when subtracting a linear expression with more than one term.
 - For example, $3 - (2x + 7)$ may be incorrectly rewritten as $3 - 2x + 7$. To address this misconception, use algebra tiles or other manipulatives to physically show the removal or subtraction of an expression.

Strategies to Support Tiered Instruction

- Teacher provides instruction to students that may incorrectly add and subtract unlike terms by giving students examples of like and unlike expressions and having students decide if they are equivalent. Give students the opportunity to explain why the expressions are equivalent or not equivalent.
- Teacher provides instruction on using the properties of operations to group like terms together.
- Instruction includes the use of different color highlighters or shapes to identify algebraic terms and constants and then combining like terms identified with the same color.
 - For example, the expression $3x + 5 + \frac{2}{3}x - 6$ can be color-coded as $3x + 5 + \frac{2}{3}x - 6$ to determine that $3x + \frac{2}{3}x + 5 - 6 = 3\frac{2}{3}x - 1$.
- Teacher includes a leading coefficient of 1 in front of grouping symbols to help students recognize the implications of a negative sign in front of grouping symbols.
 - For example, $\frac{1}{3} - (2x + \frac{7}{3})$ could be written as $\frac{1}{3} + (-1)(2x + \frac{7}{3})$ or as $\frac{1}{3} + (-1) \cdot (2x + \frac{7}{3})$ to help remind students to distribute the negative one before performing the next operation.
- Teacher provides students with a visual picture to decide if expressions are equivalent.
 - For example, if s represents the number of blue squares and t represents the number of red triangles shown, students can write an expression to represent the display. Students could write the expression as $s + s + s + t + t$, and then rewrite with combining like terms as $3s + 2t$.



- Instruction includes building a foundation for the properties of operations by modeling the associative, commutative, and distributive properties with algebra tiles for numeric and algebraic expressions and allowing the students to manipulate the algebra tiles as well.

- Instruction includes providing students with two different linear expressions already represented with algebra tiles and then allowing the students to add and subtract the algebra tiles to determine their sum or difference.
- Teacher co-creates a graphic organizer with students to review operations with positive fractions and operations with integers to assist when applying operations with rational numbers.
- Teacher provides a sorting activity to organize terms into groups before performing operations with those terms. Have students verbalize the criteria for being considered like terms to ensure they are focused on the correct similarities.
- Instruction includes using algebra tiles or other manipulatives to physically show the removal or subtraction of an expression.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Part A. Write three expressions that are equivalent to $\frac{3}{4}x - 1$.

Part B. Compare the expressions from Part A with a partner.

- How many are the same?
- How many are different?
- How many are correct?
- If there are incorrect expressions, what were the errors? Explain your reasoning.

Instructional Items

Instructional Item 1

Write the following expression using the fewest possible terms.

$$(5x - 1) - (0.06x - 4 + 0.4x)$$

Instructional Item 2

Write the following expression using the fewest possible terms.

$$\left(4 + \frac{2}{3}x\right) + \left(\frac{1}{6}x - 3\right)$$

Instructional Item 3

Write the following expression using the fewest possible terms.

$$\left(2x - \frac{4}{5} - 3z\right) - (-8x + 2z)$$

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*